



Building Quarkus Extensions

A Practical Guide for Java Developers

May 28, 2026

Agenda

- What is Quarkus
- Architecture
- Extension API
- Building a Quarkus Extension

About



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What is Quarkus

- Full-stack, cloud-native Java framework
- Runs with both JIT and AOT (native compilation)
- Optimizes Java applications for containers
- Targets serverless, cloud, Kubernetes etc.

What is Quarkus

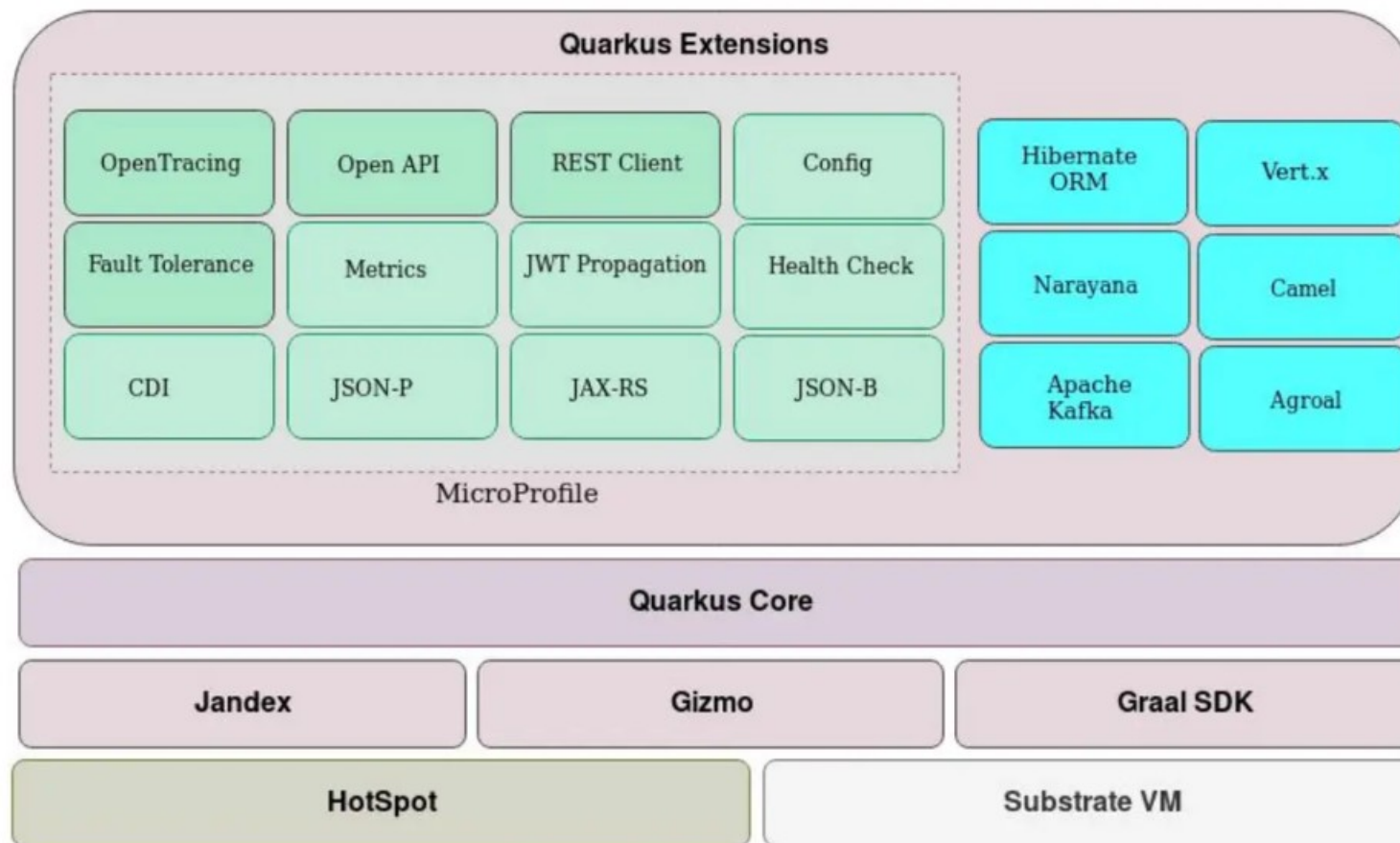
- Initial release – March 2019
- Initially developed by RedHead
- 1.0 – November 2019
- 3.36 – May 2026

Features

- Container First
- Reactive Core
- Community and Standards
- Kubernetes-Native
- High Performance
- Developer Joy

<https://quarkus.io/>

Architecture

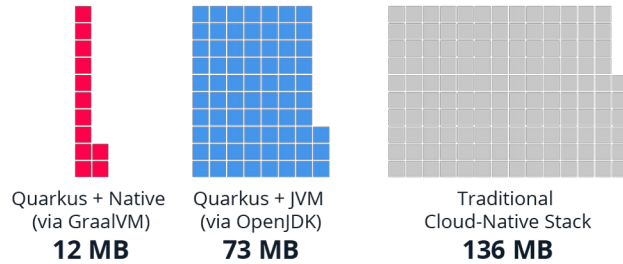


Performance

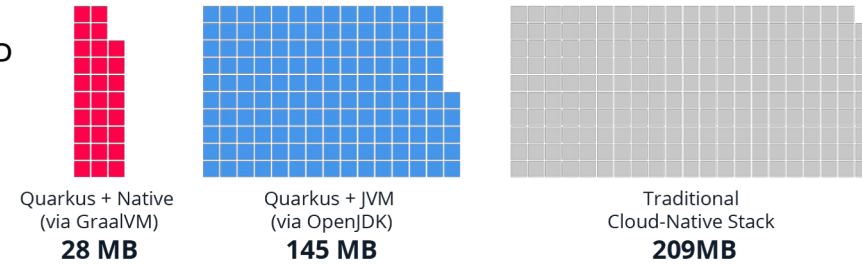
Memory (RSS) in Megabytes*

*Tested on a single-core machine

REST



REST + CRUD



BOOT + First Response Time

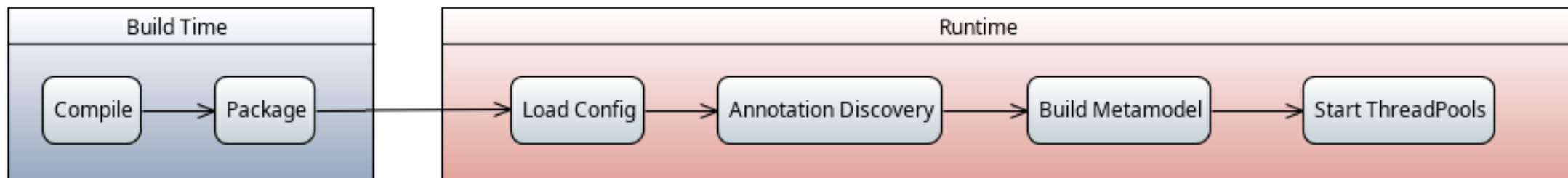
REST



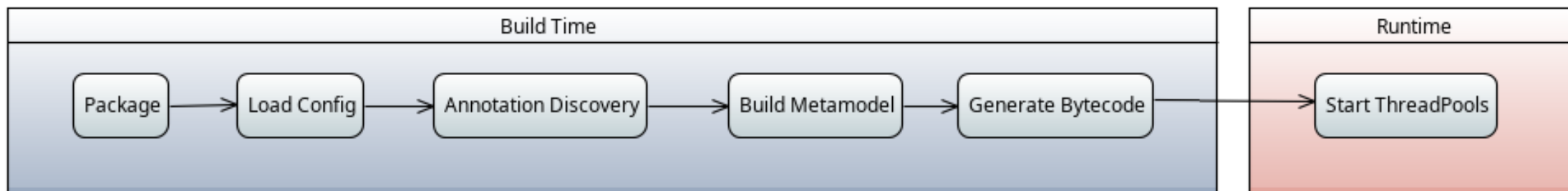
REST + CRUD



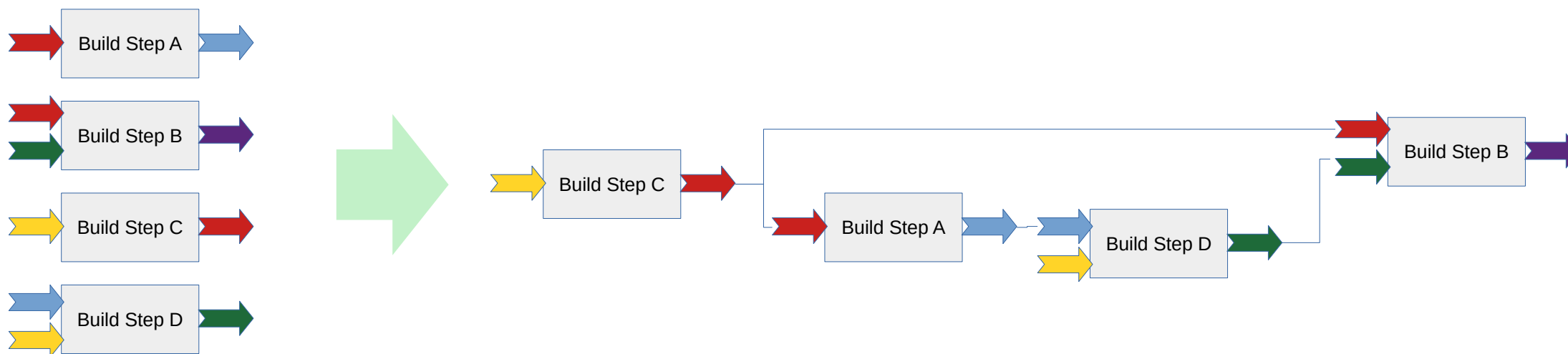
Regular app bootstrap lifecycle



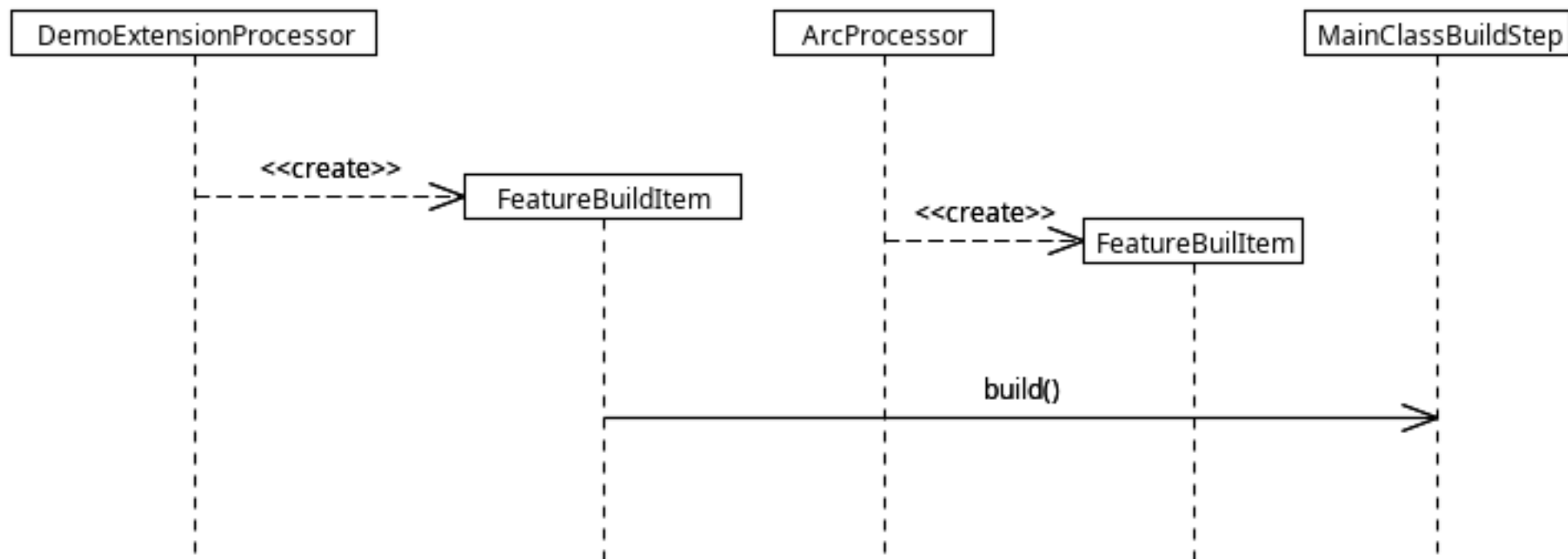
Quarkus app bootstrap lifecycle



Quarkus Build Flow



Quarkus Build Flow



Extension descriptors

quarkus-extensions.properties

```
deployment-artifact=my.demo.jconslovenia2026.quarkus\:demo-extension-  
deployment\:1.0.0-SNAPSHOT  
requires-quarkus-version=[3.36, )
```

quarkus-extensions.yaml

```
---  
name: "Demo Extension"  
metadata:  
  built-with-quarkus-core: "3.36.0"  
  requires-quarkus-core: "[3.36, )"  
  minimum-java-version: "25"  
  extension-dependencies:  
    - "io.quarkus:quarkus-arc"  
    - "io.quarkus:quarkus-core"  
artifact: "my.demo.jconslovenia2026.quarkus:demo-extension::jar:1.0.0-SNAPSHOT"
```

Extension config

```
@ConfigRoot(phase = ConfigPhase.BUILD_TIME)
@ConfigMapping(prefix = "quarkus.demo")
public interface DemoConfig {
    /**
     * Path to demo endpoint.
     *
     * @return path
     */
    @WithDefault("/demo")
    String path();
}
```

Build Step

```
@BuildStep
@Record(ExecutionTime.RUNTIME_INIT)
RouteBuildItem createRoute(CurateOutcomeBuildItem curateOutcomeBuildItem,
                           ApplicationInfoBuildItem infoApplication,
                           DemoRecorder recorder,
                           DemoConfig config) throws
    JsonProcessingException {
    ...
}
```

Build Items

- Build items which exist for existing Quarkus project extensions:
<https://quarkus.io/guides/all-builditems>
- The build items used for the demo:
 - `io.quarkus.deployment.pkg.builditem.CurateOutcomeBuildItem`
 - `io.quarkus.deployment.builditem.ApplicationInfoBuildItem`
- New custom build items could be created when needed

The Demo Extension project repo

- The “target” project repo could be found at
<https://github.com/vepeter/demo-quarkus-extension-jcon-slovenia-2026>

Feedback



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